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Section 1: System of Linear Equations

```
disp('Section 1, Exercise 1.1.8');

R = [
    1  1  5   0;
    0  1  9   0;
    0  0  7  -7
];

A = R(:, 1:3);
b = R(:, 4);

x = A \ b;

disp('Solution vector [x; y; z] = ');
disp(x);
```

```
Section 1, Exercise 1.1.8
Solution vector [x; y; z] =
    -4
     9
    -1
```

Section 2: The Matrix Equation Ax = b

```
disp('Section 2, Exercise 1.4.9');

A2 = [4 1 -7;
      0 1 6];
b2 = [8; 0];

syms x1 x2 x3;
x_vec = [x1; x2; x3];

disp('Vector equation A2 * x_vec = b2 yields:');
disp(A2 * x_vec);
```

```
disp('Right-hand side vector b2 = ');
disp(b2);

disp('Matrix equation:');
disp('    [4  1 -7;');
disp('     0  1  6] * [x1; x2; x3] = [8; 0]');
```

```
Section 2, Exercise 1.4.9
Vector equation A2 * x_vec = b2 yields:
4*x1 + x2 - 7*x3
       x2 + 6*x3

Right-hand side vector b2 =
     8
     0

Matrix equation:
    [4  1 -7;
     0  1  6] * [x1; x2; x3] = [8; 0]
```

Section 3: Matrix Operations

```
disp('Section 3, Exercise 2.1.4');

A3 = [9  -1  3;
```

```
      -8  7 -3;
      -4  1  8];

I3 = 5 * eye(3);

disp('A - 5*I3 =');
disp(A3 - I3);

disp('(5*I3) * A =');
disp(I3 * A3);
```

```
Section 3, Exercise 2.1.4
A - 5*I3 =
     4    -1     3
    -8     2    -3
    -4     1     3

(5*I3) * A =
    45    -5    15
   -40    35   -15
   -20     5    40
```

Section 4: Subspaces of Rn

```
disp('Section 4, Exercise 2.8.31');

A4 = [4  5  9  -2;
      6  5  1  12;
      3  4  8  -3];

[R4, pivcols] = rref(A4);

disp('Pivot columns of A4:'); disp(pivcols);
basis_ColA = A4(:, pivcols);
disp('Basis for Col A:'); disp(basis_ColA);

basis_NulA = null(A4, 'r');
disp('Basis for Nul A:'); disp(basis_NulA);
```

```
Section 4, Exercise 2.8.31
Pivot columns of A4:
     1     2

Basis for Col A:
     4     5
     6     5
     3     4

Basis for Nul A:
     4    -7
    -5     6
     1     0
```

```
     0     1
```

Section 5: Properties of Determinants

```
disp('Section 5, Exercise 3.2.35');

B = [1 0 1;
      2 4 5;
      3 5 6];

B_power4 = B^4;

disp('B^4 =');
disp(det(B_power4));
```

```
Section 5, Exercise 3.2.35
B^4 =
   81.0000
```

Section 6: Null Spaces, Column Spaces, Row Spaces, and LinearTransformations

```
disp('Section 6, Exercise 4.2.3');

A6 = [1 3 5 0;
      0 1 4 -2];

basis_NulA6 = null(A6, 'r');

disp('Basis vectors for Nul A:');
disp(basis_NulA6);
```

```
Section 6, Exercise 4.2.3
Basis vectors for Nul A:
     7    -6
    -4     2
     1     0
     0     1
```

Section 7: Change of Basis

```
disp('Section 7, Exercise 4.6.15');

P = [1   3   0;
    -2  -4   2;
     1   4   3];

disp('Change-of-coordinates matrix P (from B-coords to standard coords):');
```

```
disp(P);

p_C = [1; 2; 0];

x_B = P \ p_C;

disp('B-coordinate vector for p(t)=1+2t:');
disp(x_B);
```

```
Section 7, Exercise 4.6.15
Change-of-coordinates matrix P (from B-coords to standard coords):
     1     3     0
    -2    -4     2
     1     4     3

B-coordinate vector for p(t)=1+2t:
   -9.5000
    3.5000
   -1.5000
```

Section 8: Orthogonal Sets

```
disp('Section 8, Exercise 6.2.11');

u = [1; 7];
v = [-4; 2];

proj_uv = (dot(u, v) / dot(v, v)) * v;

disp('Orthogonal projection:');
disp(proj_uv);
```

```
Section 8, Exercise 6.2.11
Orthogonal projection:
    -2
     1
```

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